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Integrated Circuitry Comprising a Memory Array Comprising Strings of Memory Cells and Methods Used in Forming a Memory Array Comprising Strings of Memory Cells

Abstract

A method used in forming a memory array comprising strings of memory cells comprises forming an upper stack directly above a lower stack. The lower stack comprises vertically-alternating lower-first-tiers and lower-second-tiers. The upper stack comprises vertically-alternating upper-first-tiers and upper-second-tiers. Lower channel openings extend through the lower-first-tiers and the lower-second-tiers. The lower channel openings have sacrificial material therein. An upper of the lower-first-tiers or a lower of the upper-first-tiers comprises non-stoichiometric silicon nitride comprising (a) or (b), where (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33. A higher of the upper-first-tiers that is above said lower upper-first-tier comprises silicon nitride not having either the (a) or the (b). Upper channel openings are etched through the upper-first-tiers and the upper-second-tiers to stop on said upper lower-first-tier or said lower upper-first-tier. After the stop, the sacrificial material is removed from the lower channel openings and form channel-material strings in the upper and lower channel openings. Other embodiments, including structure independent of method, are disclosed.

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Background/Summary

RELATED PATENT DATA [0001] This patent resulted from a continuation application of U.S. patent application Ser. No. 18/244,169, filed Sep. 8, 2023, which resulted from a divisional application of U.S. patent application Ser. No. 17/068,430, filed Oct. 12, 2020, now U.S. Pat. No. 11,792,983, entitled “Integrated Circuitry Comprising A Memory Array Comprising Strings Of Memory Cells And Methods Used In Forming A Memory Array Comprising Strings Of Memory Cells”, naming Daniel Billingsley, Jordan D. Greenlee, John D. Hopkins, Yongjun Jeff Hu, and Swapnil Lengade as inventors, which claims benefit of U.S. Provisional Patent Application Ser. No. 63/071,506, filed Aug. 28, 2020, entitled “Integrated Circuitry Comprising A Memory Array Comprising Strings Of Memory Cells And Methods Used In Forming A Memory Array Comprising Strings Of Memory Cells”, naming Daniel Billingsley, Jordan D. Greenlee, John D. Hopkins, Yongjun Jeff Hu, and Swapnil Lengade as inventors, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] Embodiments disclosed herein pertain to integrated circuitry comprising a memory array comprising strings of memory cells and to methods used in forming a memory array comprising strings of memory cells.

BACKGROUND

[0003] Memory is one type of integrated circuitry and is used in computer systems for storing data. Memory may be fabricated in one or more arrays of individual memory cells. Memory cells may be written to, or read from, using digitlines (which may also be referred to as bitlines, data lines, or sense lines) and access lines (which may also be referred to as wordlines). The sense lines may conductively interconnect memory cells along columns of the array, and the access lines may conductively interconnect memory cells along rows of the array. Each memory cell may be uniquely addressed through the combination of a sense line and an access line.

[0004] Memory cells may be volatile, semi-volatile, or non-volatile. Non-volatile memory cells can store data for extended periods of time in the absence of power. Non-volatile memory is conventionally specified to be memory having a retention time of at least about 10 years. Volatile memory dissipates and is therefore refreshed/rewritten to maintain data storage. Volatile memory may have a retention time of milliseconds or less. Regardless, memory cells are configured to retain or store memory in at least two different selectable states. In a binary system, the states are

considered as either a “0” or a “1”. In other systems, at least some individual memory cells may be configured to store more than two levels or states of information.

[0005] A field effect transistor is one type of electronic component that may be used in a memory cell. These transistors comprise a pair of conductive source/drain regions having a semiconductive channel region there-between. A conductive gate is adjacent the channel region and separated therefrom by a thin gate insulator. Application of a suitable voltage to the gate allows current to flow from one of the source/drain regions to the other through the channel region. When the voltage is removed from the gate, current is largely prevented from flowing through the channel region. Field effect transistors may also include additional structure, for example a reversibly programmable charge-storage region as part of the gate construction between the gate insulator and the conductive gate.

[0006] Flash memory is one type of memory and has numerous uses in modern computers and devices. For instance, modern personal computers may have BIOS stored on a flash memory chip. As another example, it is becoming increasingly common for computers and other devices to utilize flash memory in solid state drives to replace conventional hard drives. As yet another example, flash memory is popular in wireless electronic devices because it enables manufacturers to support new communication protocols as they become standardized, and to provide the ability to remotely upgrade the devices for enhanced features.

[0007] NAND may be a basic architecture of integrated flash memory. A NAND cell unit comprises at least one selecting device coupled in series to a serial combination of memory cells (with the serial combination commonly being referred to as a NAND string). NAND architecture may be configured in a three-dimensional arrangement comprising vertically-stacked memory cells individually comprising a reversibly programmable vertical transistor. Control or other circuitry may be formed below the vertically-stacked memory cells. Other volatile or non-volatile memory array architectures may also comprise vertically-stacked memory cells that individually comprise a transistor.

[0008] Memory arrays may be arranged in memory pages, memory blocks and partial blocks (e.g., sub-blocks), and memory planes, for example as shown and described in any of U.S. Patent Application Publication Nos. 2015/0228651, 2016/0267984, and 2017/0140833. The memory blocks may at least in part define longitudinal outlines of individual wordlines in individual wordline tiers of vertically-stacked memory cells. Connections to these wordlines may occur in a so-called “stair-step structure” at an end or edge of an array of the vertically-stacked memory cells. The stair-step structure includes individual “stairs” (alternately termed “steps” or “stair-steps”) that define contact regions of the individual wordlines upon which elevationally-extending conductive vias contact to provide electrical access to the wordlines.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagrammatic cross-sectional view of a portion of a substrate in process in accordance with an embodiment of the invention and is taken through line 1-1 in FIG. 2.

[0010] FIG. 2 is a diagrammatic cross-sectional view taken through line 2-2 in FIG. 1.

[0011] FIG. 3 is an enlarged view of a portion of FIGS. 1 and 2.

[0012] FIGS. 4-7 and 10-26 are diagrammatic sequential sectional, expanded, enlarged, and/or partial views of the construction of FIGS. 1-3, or portions thereof, in process in accordance with some embodiments of the invention.

[0013] FIGS. 8 and 9 show an alternate example method and/or structural embodiment of the invention.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0014] Embodiments of the invention encompass methods used in forming a memory array, for example an array of NAND or other memory cells that may have at least some peripheral control circuitry under the array (e.g., CMOS-under-array). Embodiments of the invention encompass so-called “gate-last” or “replacement-gate” processing, so-called “gate-first” processing, and other processing whether existing or future-developed independent of when transistor gates are formed. Embodiments of the invention also encompass a memory array (e.g., NAND architecture) independent of method of manufacture. First example method embodiments are described with reference to FIGS. **1-26** which may be considered as “gate-last” or “replacement-gate” processing, and starting with FIGS. **1-3**.

[0015] FIGS. **1-3** show a construction **10** having an array or array area **12** in which elevationally-extending strings of transistors and/or memory cells will be formed. Construction **10** comprises a base substrate **11** having any one or more of conductive/conductor/conducting, semiconductive/semiconductor/semiconducting, or insulative/insulator/insulating (i.e., electrically herein) materials. Various materials have been formed elevationally over base substrate **11**. Materials may be aside, elevationally inward, or elevationally outward of the FIGS. **1-3**-depicted materials. For example, other partially or wholly fabricated components of integrated circuitry may be provided somewhere above, about, or within base substrate **11**. Control and/or other peripheral circuitry for operating components within an array (e.g., array **12**) of elevationally-extending strings of memory cells may also be fabricated and may or may not be wholly or partially within an array or sub-array. Further, multiple sub-arrays may also be fabricated and operated independently, in tandem, or otherwise relative one another. In this document, a “sub-array” may also be considered as an array.

[0016] A conductor tier **16** comprising conductor material **17** (e.g., conductively-doped polysilicon atop WSi.sub.x) has been formed above substrate **11**. Conductor tier **16** may comprise part of control circuitry (e.g., peripheral-under-array circuitry and/or a common source line or plate) used to control read and write access to the transistors and/or memory cells that will be formed within array **12**.

[0017] A lower stack **18L** comprising vertically-alternating lower-insulative-tiers **20L*** and lower-conductive-tiers **22L*** has been formed above conductor tier **16** (an * being used as a suffix to be inclusive of all such same-numerically-designated components that may or may not have other suffixes). Example thickness for each of lower tiers **20L*** and **22L*** is 22 to 60 nanometers. Only a small number of lower tiers **20L*** and **22L*** is shown, with more likely lower stack **18L** comprising dozens, a hundred or more, etc. of lower tiers **20L*** and **22L***. Other circuitry that may or may not be part of peripheral and/or control circuitry may be between conductor tier **16** and lower stack **18L**. For example, multiple vertically-alternating tiers of conductive material and insulative material of such circuitry may be below a lowest of lower-conductive-tiers **22L*** and/or above an uppermost of lower-conductive-tiers **22L***. For example, one or more select gate tiers (not shown) or dummy tiers (not shown) may be between conductor tier **16** and the lowest-conductive-tier **22L*** and one or more select gate tiers (not shown) or dummy tiers (not shown) may be above an uppermost of lower-conductive-tiers **22L***. Alternately or additionally, at least one of the depicted lowest conductive tiers **22L*** may be a select gate tier. Regardless, lower-conductive-tiers **22L*** (alternately referred to as lower-first-tiers) may not comprise conducting material and lower-insulative-tiers **20L*** (alternately referred to as lower-second-tiers) may not comprise insulative material or be insulative at this point in processing in conjunction with the hereby initially-described example method embodiment which is “gate-last” or “replacement-gate”. Example lower-conductive-tiers **22L*** comprise first material **26** (e.g., that comprises silicon nitride as further explained below) which may be wholly or partially sacrificial. Example lower-insulative-tiers **20L*** comprise second material **24** (e.g., silicon dioxide) that is of different composition from that of first material **26** and which may be wholly or partially sacrificial. For purposes of the continuing discussion, lower-first-tiers **22L*** may be considered as comprising an upper lower-first-

tier **22LU** which in one embodiment as shown is the uppermost of lower-first-tiers **22L***. Lower stack **18L** may have an uppermost tier of a lowest tier that is either a lower-first-tier **22L*** or a lower-second-tier **20L***.

[0018] Lower channel openings **25** have been formed (e.g., by etching) through lower-insulative-tiers **20L*** and lower-conductive-tiers **22L*** to conductor tier **16**. Lower channel openings **25** may taper radially-inward (not shown) moving deeper into lower stack **18L**. In some embodiments, lower channel openings **25** may go into conductor material **17** of conductor tier **16** as shown or may stop there-atop (not shown). Alternately, as an example, lower channel openings **25** may stop atop or within the lowest lower-insulative-tier **20L***. A reason for extending lower channel openings **25** at least to into conductor material **17** of conductor tier **16** is to provide an anchoring effect to material that is within lower channel openings **25**. Etch-stop material (not shown) may be within or atop conductor material **17** of conductor tier **16** to facilitate stopping of the etching of lower channel openings **25** relative to conductor tier **16** when such is desired. Such etch-stop material may be sacrificial or non-sacrificial. Regardless, lower channel openings **25** may be considered as having an average longitudinal axis **75** (e.g., average if axis **75** is not perfectly straight) that in one embodiment is vertical.

[0019] Horizontally-elongated lower trenches **40L** have been formed (e.g., by anisotropic etching) into lower stack **18L** to form laterally-spaced memory-block regions **58**. By way of example and for brevity only, lower channel openings **25** are shown as being arranged in groups or columns of staggered rows of four and five lower channel openings **25** per row and being arrayed in laterally-spaced memory-block regions **58** that will comprise laterally-spaced memory blocks **58** in a finished circuitry construction. In this document, “block” is generic to include “sub-block”. Lower trenches **40L** will typically be wider than lower channel openings **25** (e.g., 10 to 20 times wider, yet such wider degree not being shown for brevity). Memory-block regions **58** and resultant memory blocks **58** (not yet shown) may be considered as being longitudinally elongated and oriented, for example along a direction **55**. Any alternate existing or future-developed arrangement and construction may be used.

[0020] Sacrificial material **59** has been formed in lower channel openings **25** in lower-first-tiers **22L*** and in lower-second-tiers **20L***. In one embodiment and as shown, sacrificial material **59** has been formed in lower trenches **40L**. In one embodiment, sacrificial material **59** in lower channel openings **25** comprises a radially-outer silicon dioxide **70**, a radially-inner silicon dioxide **72**, and aluminum oxide **71** radially there-between. Lower trenches **40L** may comprise corresponding aluminum oxide **71** and silicon dioxide **70**, **72**. Materials **70**, **71**, and **72** are only so designated in FIG. **3** for clarity in other figures. In one embodiment, radially-outer silicon dioxide **70** and radially-inner silicon dioxide **72** are of different composition relative one another, and in one such embodiment the different composition is characterized by concentration of at least one of boron and phosphorus. As but one specific example, radially-inner silicon dioxide **72** is BPSG and radially-outer silicon dioxide **70** is undoped silicon dioxide.

[0021] Referring to FIG. **4**, an upper stack **18U** comprising vertically-alternating upper-insulative-tiers **20U*** (alternately referred to as upper-second-tiers) and upper-conductive-tiers **22U*** (alternately referred to as upper-first-tiers) has been formed above lower stack **18L**. Upper-insulative-tiers **20U*** and upper-conductive-tiers **22U*** may have any of the attributes described above with respect to lower-insulative-tiers **20L*** and lower-conductive-tiers **22L***. Example upper-conductive-tiers **22U*** comprise first material **26** (e.g., that comprises silicon nitride as further explained below) which may be wholly or partially sacrificial. Example upper-insulative-tiers **20U*** are shown as comprising second material **24** and upper-conductive-tiers **22U*** are shown as comprising first material **26**, although other compositions may of course be used and not necessarily of the same composition as in lower stack **18L**. For purposes of the continuing discussion, upper-first-tiers **22U*** may be considered as comprising a lower upper-first-tier **22UL** which in one embodiment as shown is the lowest of upper-first-tiers **22U***. Upper stack **18U** may

have an uppermost tier or a lowest tier that is either an upper-first-tier **22U*** or an upper-second-tier **20U***.

[0022] An upper of lower-first-tiers **22L*** or a lower of upper-first-tiers **22U*** comprises non-stoichiometric silicon nitride comprising (a) or (b), where: [0023] (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and [0024] (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33;

A higher upper-first-tier **22U*** that is above the lower upper-first-tier (regardless of composition of said lower upper-first-tier) comprises silicon nitride not having either the (a) or the (b). Such higher upper-first-tier **22U*** may be stoichiometric (i.e., nitrogen-to-silicon atomic ratio equal to 1.33) or may be non-stoichiometric (e.g., nitrogen-to-silicon atomic ratio of at least 1.5).

[0025] In one embodiment, the upper of lower-first-tiers **22L*** is that which comprises the non-stoichiometric silicon nitride and in one such embodiment is uppermost lower-first-tier **22LU**. Such is exemplified in the figures by different stippling in material **26** that comprises the non-stoichiometric silicon nitride comprising the (a) or the (b) in comparison to material **26** that comprises silicon nitride not having either the (a) or the (b). Regardless, in one embodiment, multiple upper lower-first-tiers **22L*** comprise the non-stoichiometric silicon nitride (e.g., one or more tiers **22L** below uppermost lower-first-tier **22LU** [not shown] and that may or may not include uppermost lower-first-tier **22LU**). In one embodiment, the lower of upper-first-tiers **22U*** is that which comprises the non-stoichiometric silicon nitride (not shown) and in one such embodiment is lower upper-first-tier **22UL**. Regardless, in one embodiment, multiple lower upper-first-tiers **22U*** comprise the non-stoichiometric silicon nitride (e.g., one or more tiers **22U** above lower upper-first-tier **22UL** [not shown] and that may or may not include lower upper-first-tier **22UL**). In one embodiment, each of the upper of lower-first-tiers **22L*** and the lower of upper-first-tiers **22U*** comprises the non-stoichiometric silicon nitride (and that may include any of the immediately-above-stated attributes).

[0026] In one embodiment, the non-stoichiometric silicon nitride comprises the (a), and in one such embodiment the nitrogen-to-silicon atomic ratio is 1.35 to 1.48. In one embodiment, the non-stoichiometric silicon nitride comprises the (b), and in one such embodiment the nitrogen-to-silicon atomic ratio is 1.2 to 1.3. In one embodiment, some of the non-stoichiometric silicon nitride comprises the (a) and another some of the non-stoichiometric silicon nitride comprises the (b).

[0027] Upper channel openings **39** are etched through upper-first-tiers **22U*** and upper-second-tiers **20U*** to stop on the upper lower-first-tier or the lower upper-first-tier that comprises the non-stoichiometric silicon nitride comprising the (a) or the (b). FIGS. **5** and **6** show upper channel openings **39** as having been etched through upper-first-tiers **22U*** and upper-second-tiers **20U*** (and through an uppermost tier **20L***) to stop on upper lower-first-tier **22LU** (i.e., atop or within) which in this example comprises the non-stoichiometric silicon nitride comprising the (a) or the (b). In one such embodiment and as shown, the etching of upper channel openings **39** to the stop (i.e., the stopping of the etching using the non-stoichiometric silicon nitride comprising the (a) or the (b) of tier **22LU** as an etch stop) exposes sacrificial material **59**. In one embodiment and as shown, and as may be best perceived with reference to FIG. **7**, individual upper channel openings **39** are formed to have an average longitudinal axis **85** that is laterally offset relative to lower-portion average longitudinal axis **75** in a vertical cross-section (e.g. that of FIG. **6**) where upper channel openings **39** and lower channel openings **25** join. Alternately, average longitudinal axis **85** may be angled (at other than the straight angle) relative to average longitudinal axis **75**.

[0028] An alternate example is shown in FIGS. **8** and **9** with respect to a construction **10a**. Like numerals from the above-described embodiments have been used where appropriate, with some construction differences being indicated with the suffix “a” or with different numerals. FIG. **8** shows etching upper channel openings **39** through upper-first-tiers **22U*** and upper-second-tiers **20U*** to stop on lower upper-first-tier **22UL** (i.e., atop or within) which in this example comprises the non-stoichiometric silicon nitride comprising the (a) or the (b). Thereby, the etching of upper

channel openings **39** to the stop (i.e., the stopping of the etching using the non-stoichiometric silicon nitride comprising the (a) or the (b) of tier **22UL** as an etch stop) does not expose sacrificial material **59**. FIG. **9** shows subsequent etching through lower upper-first-tier **22UL** to expose sacrificial material **59**. Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0029] Referring to FIGS. **10** and **11**, and back to construction **10**, sacrificial material **59** (i.e., at least some) has been removed (e.g., by etching) from individual lower channel openings **25** through upper channel openings **39**. Where, for example, materials **70**, **71**, and **72** are present, some of material(s) **70** and/or **71** may remain (not shown) in lower channel openings **25** to facilitate formation of transistor material (described below).

[0030] Transistor channel material may be formed in the individual channel openings elevationally along the insulative tiers and the conductive tiers, thus comprising individual channel-material strings, which is directly electrically coupled with conductive material in the conductor tier. Individual memory cells of the example memory array being formed may comprise a gate region (e.g., a control-gate region) and a memory structure laterally-between the gate region and the channel material. In one such embodiment, the memory structure is formed to comprise a charge-blocking region, storage material (e.g., charge-storage material), and an insulative charge-passage material. The storage material (e.g., floating gate material such as doped or undoped silicon or charge-trapping material such as silicon nitride, metal dots, etc.) of the individual memory cells is elevationally along individual of the charge-blocking regions. The insulative charge-passage material (e.g., a band gap-engineered structure having nitrogen-containing material [e.g., silicon nitride] sandwiched between two insulator oxides [e.g., silicon dioxide]) is laterally-between the channel material and the storage material.

[0031] FIGS. **12-15** show one embodiment wherein charge-blocking material **30**, storage material **32**, and charge-passage material **34** have been formed in individual upper channel openings **39** and lower channel openings **25** elevationally along insulative tiers **20*** and conductive tiers **22***. Transistor materials **30**, **32**, and **34** (e.g., memory-cell materials) may be formed by, for example, deposition of respective thin layers thereof over upper stack **18U** and within individual openings **39** and **25** followed by planarizing such back at least to a top surface of upper stack **18U**.

[0032] Channel material **36** has also been formed in channel openings **39/25** elevationally along insulative tiers **20U*/20L*** and conductive tiers **22U*/22L***, thus comprising individual operative channel-material strings **53** in channel openings **39/25**. Channel-material strings **53** in one embodiment have memory-cell materials (e.g., **30**, **32**, and **34**) there-along and with second-tier material (e.g., **24**) being horizontally-between immediately-adjacent channel-material strings **53**. Materials **30**, **32**, **34**, and **36** are collectively shown as and only designated as material **37** in FIGS. **11** and **12** due to scale. Example channel materials **36** include appropriately-doped crystalline semiconductor material, such as one or more silicon, germanium, and so-called III/V semiconductor materials (e.g., GaAs, InP, GaP, and GaN). Example thickness for each of materials **30**, **32**, **34**, and **36** is 25 to 100 Angstroms. Punch etching may be conducted to remove materials **30**, **32**, and **34** from the bases of lower channel openings **25** (not shown) to expose conductor tier **16** such that channel material **36** is directly against conductor material **17** of conductor tier **16**. Such punch etching may occur separately with respect to each of materials **30**, **32**, and **34** (as shown) or may occur with respect to only some (not shown). Alternately, and by way of example only, no punch etching may be conducted and channel material **36** may be directly electrically coupled to conductor material **17** of conductor tier **16** only by a separate conductive interconnect (not yet shown). Channel openings **39/25** are shown as comprising a radially-central solid dielectric material **38** (e.g., spin-on-dielectric, silicon dioxide, and/or silicon nitride). Alternately, and by way of example only, the radially-central portion within channel openings **39/25** may include void space(s) (not shown) and/or be devoid of solid material (not shown).

[0033] Referring to FIGS. **16** and **17**, horizontally-elongated upper trenches **40U** have been formed

to lower trenches **40L** and sacrificial material **59** (not shown) has then been removed therefrom (e.g., by selective etching).

[0034] Referring to FIGS. **18** and **19**, first-tier material **26** in first tiers **22*** has been isotropically etched through trenches **40U/40L** selectively relative to second-tier material **24** in second tiers **20*** (e.g., using liquid or vapor H.sub.3PO.sub.4 as a primary etchant where material **26** comprises silicon nitride and exposed other materials comprise one or more oxides or polysilicon). In one embodiment and as shown, upper lower-first-tier **22LU** comprises non-stoichiometric silicon nitride **26** comprising the (a) or the (b). Such may be more resistant to the etching chemistry that etches silicon nitride **26** that does not have either the (a) or the (b). In such instance, and in one embodiment as shown, insulative material **26** comprising the non-stoichiometric silicon nitride comprising the (a) or the (b) may remain about some channel-material strings **53** (including about materials **30**, **32**, and **34** when present) in at least one of conductive/first tiers **22*** (e.g., upper lower-first-tier **22LU** in this example). In one embodiment and as shown, insulative material **26** comprising the non-stoichiometric silicon nitride comprising the (a) or the (b) (e.g. upper lower-first-tier **22LU**) comprises a continuous horizontally-elongated insulative line **69** in and horizontally along memory-block region **58**.

[0035] Referring to FIGS. **20-26**, conducting material **48** has been deposited into trenches **40U/40L** to fill volume in conductive tiers **22*** the result of removing material **26**. Such has thereafter been removed from trenches **40U/40L**, thus forming individual conductive lines **29** (e.g., wordlines) and elevationally-extending strings **49** of individual transistors and/or memory cells **56**.

[0036] A thin insulative liner (e.g., Al.sub.2O.sub.3 and not shown) may be formed before forming conducting material **48**. Approximate locations of transistors and/or memory cells **56** are indicated with a bracket in FIG. **24** and some with dashed outlines in FIGS. **20-23** and **25**, with transistors and/or memory cells **56** being essentially ring-like or annular in the depicted example. Alternately, transistors and/or memory cells **56** may not be completely encircling relative to individual channel openings **39/25** such that each channel opening **39/25** may have two or more elevationally-extending strings **49** (e.g., multiple transistors and/or memory cells about individual channel openings in individual conductive tiers with perhaps multiple wordlines per channel opening in individual conductive tiers, and not shown). Conducting material **48** may be considered as having terminal ends **50** (FIG. **24**) corresponding to control-gate regions **52** of individual transistors and/or memory cells **56**. Control-gate regions **52** in the depicted embodiment comprise individual portions of individual conductive lines **29**. Materials **30**, **32**, and **34** may be considered as a memory structure **65** that is laterally between control-gate region **52** and channel material **36**. In one embodiment and as shown with respect to the example “gate-last” processing, conducting material **48** of conductive tiers **22U*/22L*** is formed after forming channel openings **39/25** and/or trenches **40U/40L**. Alternately, the conducting material of the conductive tiers may be formed before forming channel openings **39/25** and/or trenches **40U/40L** (not shown), for example with respect to “gate-first” processing.

[0037] A charge-blocking region (e.g., charge-blocking material **30**) is between storage material **32** and individual control-gate regions **52**. A charge block may have the following functions in a memory cell: In a program mode, the charge block may prevent charge carriers from passing out of the storage material (e.g., floating-gate material, charge-trapping material, etc.) toward the control gate, and in an erase mode the charge block may prevent charge carriers from flowing into the storage material from the control gate. Accordingly, a charge block may function to block charge migration between the control-gate region and the storage material of individual memory cells. An example charge-blocking region as shown comprises insulator material **30**. By way of further examples, a charge-blocking region may comprise a laterally (e.g., radially) outer portion of the storage material (e.g., material **32**) where such storage material is insulative (e.g., in the absence of any different-composition material between an insulative storage material **32** and conducting material **48**). Regardless, as an additional example, an interface of a storage material and

conductive material of a control gate may be sufficient to function as a charge-blocking region in the absence of any separate-composition-insulator material **30**. Further, an interface of conducting material **48** with material **30** (when present) in combination with insulator material **30** may together function as a charge-blocking region, and as alternately or additionally may a laterally-outer region of an insulative storage material (e.g., a silicon nitride material **32**). An example material **30** is one or more of silicon hafnium oxide and silicon dioxide.

[0038] Intervening material **57** has been formed in trenches **40U/40L** and thereby laterally-between and longitudinally-along immediately-laterally-adjacent memory blocks **58**. Intervening material **57** may provide lateral electrical isolation (insulation) between immediately-laterally-adjacent memory blocks. Such may include one or more of insulative, semiconductive, and conducting materials and, regardless, may facilitate conductive tiers **22** from shorting relative one another in a finished circuitry construction. Example insulative materials are one or more of SiO₂, Si₃N₄, Al₂O₃, and undoped polysilicon. Intervening material **57** may include through array vias (not shown).

[0039] In one embodiment, not all of the pillar structures that extend through insulative tiers **20*** and conductive tiers **22*** in individual memory blocks **58** are operative. For example, and in one embodiment as shown, dummy pillars **73** (i.e., pillars that do not function as a string of memory cells) extend through insulative tiers **20*** and conductive tiers **22*** in individual memory blocks **58**. Dummy pillars **73** are spaced from one another horizontally along horizontally-elongated conductive lines **29** of conductive tiers **22***. At least one of conductive tiers **22*** (e.g., **22LU** as shown) comprises insulative material **26** therein laterally between conductive line **29** in that conductive tier **22LU** and dummy pillar **73** extending through that conductive tier **22LU**, with insulative material **26** comprising non-stoichiometric silicon nitride comprising the (a) or the (b). Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0040] In some embodiments, construction **10** may be considered as comprising a first region (e.g., as shown by FIGS. **20-22**) and a second region **70** aside the first region (e.g., as shown in FIG. **26**). Second region **70** may be laterally-contacting the first region (not shown) or may be laterally-spaced from the first region (e.g., closely laterally there-adjacent but not touching, or laterally-far there-from and not touching). Second region **70** may be within one or more of the memory blocks (not shown). In some embodiments, construction **10** may be considered as comprising a first vertical stack (e.g., stack **18*** in FIGS. **20-22**) and a second vertical stack (e.g., stack **18*** in second region **70**), with the second stack comprising an upper portion **18U** and a lower portion **18L**.

[0041] Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used in the embodiments shown and described with reference to the above embodiments.

[0042] Alternate embodiment constructions may result from method embodiments described above, or otherwise. Regardless, embodiments of the invention encompass integrated circuitry comprising a memory array independent of method of manufacture. Nevertheless, such integrated circuitry and memory array may have any of the attributes as described herein in method embodiments. Likewise, the above-described method embodiments may incorporate, form, and/or have any of the attributes described with respect to device embodiments.

[0043] In one embodiment, integrated circuitry (e.g., **10**) comprising a memory array (e.g., **12**) comprising strings (e.g., **49**) of memory cells (e.g., **56**) comprises laterally-spaced memory blocks (e.g., **58**) individually comprising a first vertical stack (e.g., **18*** in FIG. **22**) comprising alternating insulative tiers (e.g., **20***) and conductive tiers (e.g., **22***). Strings (e.g., **49**) of memory cells (e.g., **56**) comprise channel-material strings (e.g., **53**) that extend through the insulative tiers and the conductive tiers. The conductive tiers individually comprise a horizontally-elongated conductive line (e.g., **29**). A second vertical stack (e.g., **18*** in FIG. **26**) is aside the first vertical stack. The second vertical stack comprises an upper portion (e.g., **18U**) and a lower portion (e.g., **18L**). The

upper portion comprises alternating upper-first-insulating-tiers (e.g., **22U***) and upper-second-insulating-tiers (e.g., **20U***) of different composition relative one another. The lower portion comprises lower-first-insulating-tiers (e.g., **22L***) and lower-second-insulating-tiers (e.g., **20L***) of different composition relative one another. An upper of the lower-first-insulating-tiers or a lower of the upper-first-insulating-tiers comprises non-stoichiometric silicon nitride comprising (a) or (b), where: [0044] (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and [0045] (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33

A higher of the upper-first-insulating-tiers that is above said lower upper-first-insulating-tier comprises silicon nitride not having either the (a) or the (b). Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0046] In one embodiment, integrated circuitry (e.g., **10**) comprising a memory array (e.g., **12**) comprising strings (e.g., **49**) of memory cells (e.g., **56**) comprises laterally-spaced memory blocks (e.g., **58**) individually comprising a first vertical stack (e.g., **18*** in FIG. **22**) comprising alternating insulative tiers (e.g., **20***) and conductive tiers (e.g., **22***). Strings (e.g., **49**) of memory cells (e.g., **56**) comprise channel-material strings (e.g., **53**) that extend through the insulative tiers and the conductive tiers. The conductive tiers individually comprise a horizontally-elongated conductive line (e.g., **29**). Dummy pillars (e.g., **73**) extend through the insulative tiers and the conductive tiers in individual of the memory blocks. The dummy pillars are spaced from one another horizontally along the horizontally-elongated conductive lines of the conductive tiers. At least one of the conductive tiers (e.g., **22LU**) comprises insulative material (e.g., **26**) therein laterally between the conductive line in that conductive tier and the dummy pillar extending through that conductive tier. The insulative material comprises non-stoichiometric silicon nitride comprising (a) or (b), where: [0047] (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and [0048] (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33.

Any other attribute(s) or aspect(s) as shown and/or described herein with respect to other embodiments may be used.

[0049] The above processing(s) or construction(s) may be considered as being relative to an array of components formed as or within two stacks or two decks of such components above or as part of an underlying base substrate (albeit, the two stacks/decks may each have multiple tiers). Control and/or other peripheral circuitry for operating or accessing such components within an array may also be formed anywhere as part of the finished construction, and in some embodiments may be under the array (e.g., CMOS under-array). Regardless, one or more additional such stack(s)/deck(s) may be provided or fabricated above and/or below that shown in the figures or described above. Further, the array(s) of components may be the same or different relative one another in different stacks/decks and different stacks/decks may be of the same thickness or of different thicknesses relative one another. Intervening structure may be provided between immediately-vertically-adjacent stacks/decks (e.g., additional circuitry and/or dielectric layers). Also, different stacks/decks may be electrically coupled relative one another. The multiple stacks/decks may be fabricated separately and sequentially (e.g., one atop another), or two or more stacks/decks may be fabricated at essentially the same time.

[0050] The assemblies and structures discussed above may be used in integrated circuits/circuitry and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multichip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

[0051] In this document unless otherwise indicated, “elevational”, “higher”, “upper”, “lower”, “top”, “atop”, “bottom”, “above”, “below”, “under”, “beneath”, “up”, and “down” are generally with reference to the vertical direction. “Horizontal” refers to a general direction (i.e., within 10

degrees) along a primary substrate surface and may be relative to which the substrate is processed during fabrication, and vertical is a direction generally orthogonal thereto. Reference to “exactly horizontal” is the direction along the primary substrate surface (i.e., no degrees there-from) and may be relative to which the substrate is processed during fabrication. Further, “vertical” and “horizontal” as used herein are generally perpendicular directions relative one another and independent of orientation of the substrate in three-dimensional space. Additionally, “elevationally-extending” and “extend (ing) elevationally” refer to a direction that is angled away by at least 45° from exactly horizontal. Further, “extend (ing) elevationally”, “elevationally-extending”, “extend (ing) horizontally”, “horizontally-extending” and the like with respect to a field effect transistor are with reference to orientation of the transistor's channel length along which current flows in operation between the source/drain regions. For bipolar junction transistors, “extend (ing) elevationally” “elevationally-extending”, “extend (ing) horizontally”, “horizontally-extending” and the like, are with reference to orientation of the base length along which current flows in operation between the emitter and collector. In some embodiments, any component, feature, and/or region that extends elevationally extends vertically or within 10° of vertical.

[0052] Further, “directly above”, “directly below”, and “directly under” require at least some lateral overlap (i.e., horizontally) of two stated regions/materials/components relative one another. Also, use of “above” not preceded by “directly” only requires that some portion of the stated region/material/component that is above the other be elevationally outward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components). Analogously, use of “below” and “under” not preceded by “directly” only requires that some portion of the stated region/material/component that is below/under the other be elevationally inward of the other (i.e., independent of whether there is any lateral overlap of the two stated regions/materials/components).

[0053] Any of the materials, regions, and structures described herein may be homogenous or non-homogenous, and regardless may be continuous or discontinuous over any material which such overlie. Where one or more example composition(s) is/are provided for any material, that material may comprise, consist essentially of, or consist of such one or more composition(s). Further, unless otherwise stated, each material may be formed using any suitable existing or future-developed technique, with atomic layer deposition, chemical vapor deposition, physical vapor deposition, epitaxial growth, diffusion doping, and ion implanting being examples.

[0054] Additionally, “thickness” by itself (no preceding directional adjective) is defined as the mean straight-line distance through a given material or region perpendicularly from a closest surface of an immediately-adjacent material of different composition or of an immediately-adjacent region. Additionally, the various materials or regions described herein may be of substantially constant thickness or of variable thicknesses. If of variable thickness, thickness refers to average thickness unless otherwise indicated, and such material or region will have some minimum thickness and some maximum thickness due to the thickness being variable. As used herein, “different composition” only requires those portions of two stated materials or regions that may be directly against one another to be chemically and/or physically different, for example if such materials or regions are not homogenous. If the two stated materials or regions are not directly against one another, “different composition” only requires that those portions of the two stated materials or regions that are closest to one another be chemically and/or physically different if such materials or regions are not homogenous. In this document, a material, region, or structure is “directly against” another when there is at least some physical touching contact of the stated materials, regions, or structures relative one another. In contrast, “over”, “on”, “adjacent”, “along”, and “against” not preceded by “directly” encompass “directly against” as well as construction where intervening material(s), region(s), or structure(s) result(s) in no physical touching contact of the stated materials, regions, or structures relative one another.

[0055] Herein, regions-materials-components are “electrically coupled” relative one another if in

normal operation electric current is capable of continuously flowing from one to the other and does so predominately by movement of subatomic positive and/or negative charges when such are sufficiently generated. Another electronic component may be between and electrically coupled to the regions-materials-components. In contrast, when regions-materials-components are referred to as being “directly electrically coupled”, no intervening electronic component (e.g., no diode, transistor, resistor, transducer, switch, fuse, etc.) is between the directly electrically coupled regions-materials-components.

[0056] Any use of “row” and “column” in this document is for convenience in distinguishing one series or orientation of features from another series or orientation of features and along which components have been or may be formed. “Row” and “column” are used synonymously with respect to any series of regions, components, and/or features independent of function. Regardless, the rows may be straight and/or curved and/or parallel and/or not parallel relative one another, as may be the columns. Further, the rows and columns may intersect relative one another at 90° or at one or more other angles (i.e., other than the straight angle).

[0057] The composition of any of the conductive/conductor/conducting materials herein may be metal material and/or conductively-doped semiconductive/semiconductor/semiconducting material. “Metal material” is any one or combination of an elemental metal, any mixture or alloy of two or more elemental metals, and any one or more conductive metal compound(s).

[0058] Herein, any use of “selective” as to etch, etching, removing, removal, depositing, forming, and/or formation is such an act of one stated material relative to another stated material(s) so acted upon at a rate of at least 2:1 by volume. Further, any use of selectively depositing, selectively growing, or selectively forming is depositing, growing, or forming one material relative to another stated material or materials at a rate of at least 2:1 by volume for at least the first 75 Angstroms of depositing, growing, or forming.

[0059] Unless otherwise indicated, use of “or” herein encompasses either and both.

CONCLUSION

[0060] In some embodiments, a method used in forming a memory array comprising strings of memory cells comprises forming an upper stack directly above a lower stack. The lower stack comprises vertically-alternating lower-first-tiers and lower-second-tiers. The upper stack comprises vertically-alternating upper-first-tiers and upper-second-tiers. Lower channel openings extend through the lower-first-tiers and the lower-second-tiers. The lower channel openings have sacrificial material therein. An upper of the lower-first-tiers or a lower of the upper-first-tiers comprises non-stoichiometric silicon nitride comprising (a) or (b), where (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33. A higher of the upper-first-tiers that is above said lower upper-first-tier comprises silicon nitride not having either the (a) or the (b). Upper channel openings are etched through the upper-first-tiers and the upper-second-tiers to stop on said upper lower-first-tier or said lower upper-first-tier. After the stop, the sacrificial material is removed from the lower channel openings and form channel-material strings in the upper and lower channel openings.

[0061] In some embodiments, integrated circuitry comprising a memory array comprising strings of memory cell comprises laterally-spaced memory blocks individually comprising a first vertical stack comprising alternating insulative tiers and conductive tiers. Strings of memory cells comprising channel-material strings extend through the insulative tiers and the conductive tiers. The conductive tiers individually comprise a horizontally-elongated conductive line. A second vertical stack is aside the first vertical stack. The second vertical stack comprises an upper portion and a lower portion. The upper portion comprises alternating upper-first-insulating-tiers and upper-second-insulating-tiers of different composition relative one another. The lower portion comprises lower-first-insulating-tiers and lower-second-insulating-tiers of different composition relative one another. An upper of the lower-first-insulating-tiers or a lower of the upper-first-insulating-tiers

comprises non-stoichiometric silicon nitride comprising (a) or (b), where (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33. A higher of the upper-first-insulating-tiers that is above said lower upper-first-insulating-tier comprises silicon nitride not having either the (a) or the (b).

[0062] In some embodiments, integrated circuitry comprising a memory array comprising strings of memory cells comprises laterally-spaced memory blocks individually comprising a vertical stack comprising alternating insulative tiers and conductive tiers. Channel-material strings of memory cells extend through the insulative tiers and the conductive tiers. The conductive tiers individually comprise a horizontally-elongated conductive line. Dummy pillars extend through the insulative tiers and the conductive tiers in individual of the memory blocks. The dummy pillars are spaced from one another horizontally along the horizontally-elongated conductive lines of the conductive tiers. At least one of the conductive tiers comprises insulative material therein laterally between the conductive line in that conductive tier and the dummy pillar that extends through that conductive tier. The insulative material comprises non-stoichiometric silicon nitride comprising (a) or (b), where (a): a nitrogen-to-silicon atomic ratio greater than 1.33 and less than 1.5; and (b): a nitrogen-to-silicon atomic ratio greater than or equal to 1.0 and less than 1.33.

[0063] In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

Claims

1. Integrated circuitry comprising a memory array comprising strings of memory cells, comprising: a vertical stack having an upper portion and a lower portion, the upper portion comprising upper first insulating tiers alternating and in direct physical contact with upper second insulating tiers of different composition relative one another, the lower portion comprising lower first insulating tiers and lower second insulating tiers of different composition relative one another; an upper of the lower first insulating tiers or a lower of the upper first insulating tiers comprising nonstoichiometric silicon nitride comprising (a) or (b), where: (a): a nitrogen to silicon atomic ratio greater than 1.33 and less than 1.5; and (b): a nitrogen to silicon atomic ratio greater than or equal to 1.0 and less than 1.33 an upper channel opening through the upper portion; a lower channel opening through the upper portion, the lower channel opening being laterally offset relative to the upper channel opening; and channel material extending continuously through the upper and lower channel openings.
2. The integrated circuitry of claim 1 wherein the vertical stack is divided into laterally spaced memory blocks, the memory blocks comprise strings of memory cells comprise the channel material.
3. The integrated circuitry of claim 1 wherein a higher of the upper first insulating tiers that is above the lower upper first insulating tier comprises silicon nitride not having either the (a) or the (b).
4. The integrated circuitry of claim 1 wherein the nonstoichiometric silicon nitride comprises the (a).
5. The integrated circuitry of claim 4 wherein the nitrogen to silicon atomic ratio is 1.35 to 1.48.
6. The integrated circuitry of claim 1 wherein the nonstoichiometric silicon nitride comprises the (b).
7. The integrated circuitry of claim 6 wherein the nitrogen to silicon atomic ratio is 1.2 to 1.3.
8. The integrated circuitry of claim 1 wherein some of the nonstoichiometric silicon nitride

- comprises the (a) and another some of the nonstoichiometric silicon nitride comprises the (b).
- 9.** The integrated circuitry of claim 1 wherein the upper of the lower first insulating tiers comprises the nonstoichiometric silicon nitride.
- 10.** The integrated circuitry of claim 9 wherein the upper of the lower first insulating tiers is the uppermost of the lower first insulating tiers.
- 11.** The integrated circuitry of claim 9 wherein multiple of the upper lower first insulating tiers comprise the nonstoichiometric silicon nitride.
- 12.** The integrated circuitry of claim 1 wherein the lower of the upper first insulating tiers comprises the nonstoichiometric silicon nitride.
- 13.** The integrated circuitry of claim 12 wherein the lower of the upper first insulating tiers is the lowest of the upper first insulating tiers.
- 14.** The integrated circuitry of claim 12 wherein multiple of the lower upper first insulating tiers comprise the non-stoichiometric silicon nitride.
- 15.** Integrated circuitry comprising a memory array comprising strings of memory cells, comprising: laterally spaced memory blocks individually comprising a vertical stack comprising alternating insulative tiers and conductive tiers; channel material strings of memory cells extending through the insulative tiers and the conductive tiers, the conductive tiers individually comprising a horizontally elongated conductive line, the channel material strings comprising a channel material extending through the vertical stack, an upper portion of the channel material being laterally offset relative to a lower portion of the channel material; dummy pillars extending through the insulative tiers and the conductive tiers in individual of the memory blocks, the dummy pillars being spaced from one another horizontally along the horizontally elongated conductive line; and at least one of the conductive tiers comprising insulative material therein laterally between the conductive line in that conductive tier and the dummy pillar extending through that conductive tier, the insulative material comprising nonstoichiometric silicon nitride.
- 16.** The integrated circuitry of claim 15 wherein the nonstoichiometric silicon nitride comprises a nitrogen to silicon atomic ratio greater than 1.33 and less than 1.5, or a nitrogen to silicon atomic ratio greater than or equal to 1.0 and less than 1.33.
- 17.** The integrated circuitry of claim 15 wherein the insulative material comprises a continuous horizontally elongated insulative line in and horizontally along the individual memory blocks.
- 18.** The integrated circuitry of claim 15 wherein multiple of the conductive tiers comprise the nonstoichiometric silicon nitride.
- 19.** The integrated circuitry of claim 15 wherein some of the nonstoichiometric silicon nitride comprises the nitrogen to silicon atomic ratio greater than 1.33 and less than 1.5 and another some of the nonstoichiometric silicon nitride comprises the nitrogen to silicon atomic ratio greater than or equal to 1.0 and less than 1.33.
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